Graphs

April 2, 2026

1 Graphs

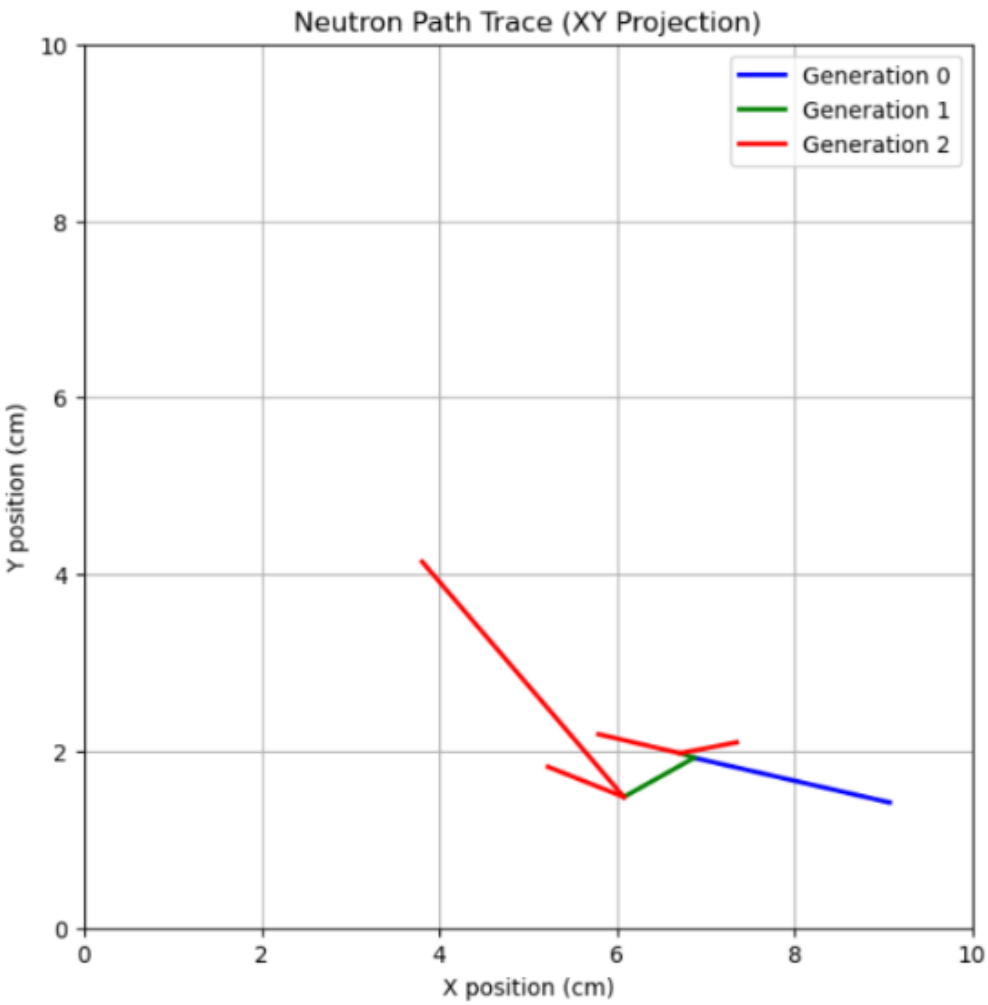


Figure 0: One plot showing the path traced by 3 neutron generations in the XY plane. Each line segment represents a neutrons movement and each color represents a generation. This graph gives us a geometrical idea of how the calculations were done, the fussion must also occur inside the chosen shape of nuclear fuel.

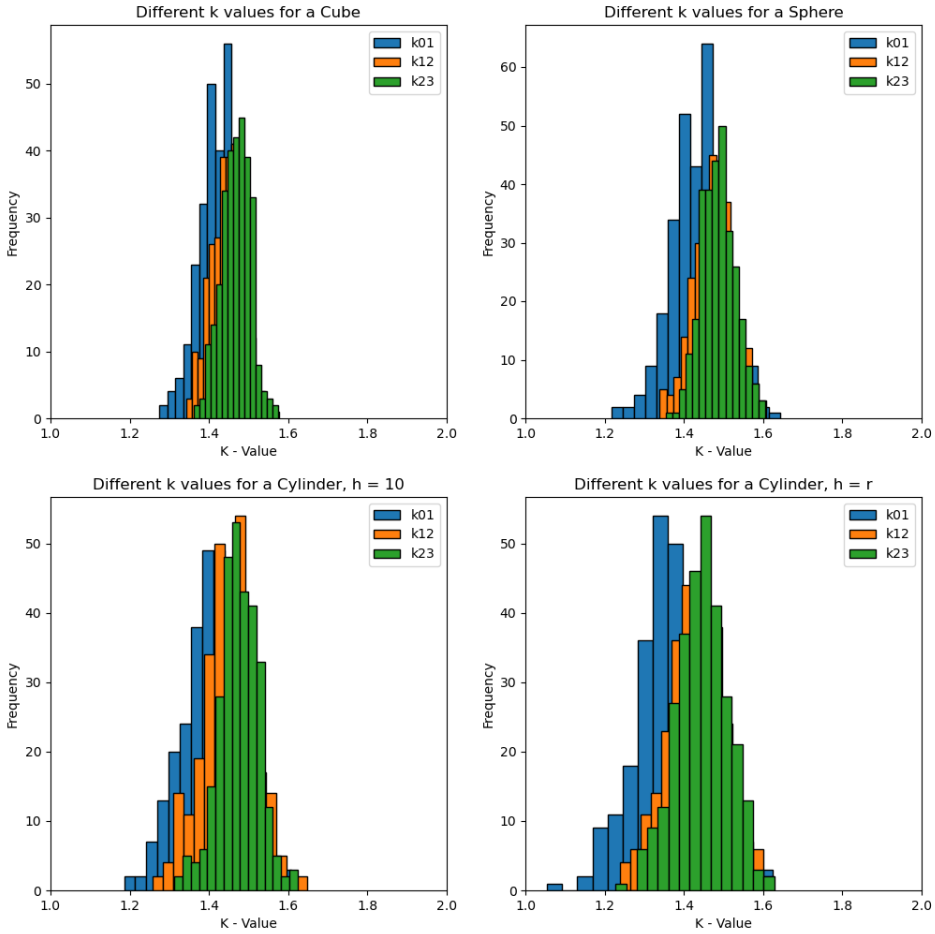


Figure 1: Four histograms showing the neutron multiplication factor k for different shapes (a cube, a sphere, and two cylinders with different heights). The different colors show how k changes from one neutron generation to the next, and the spread of the graphs comes from repeating the simulation many times to see how the shape affects neutron confinement.

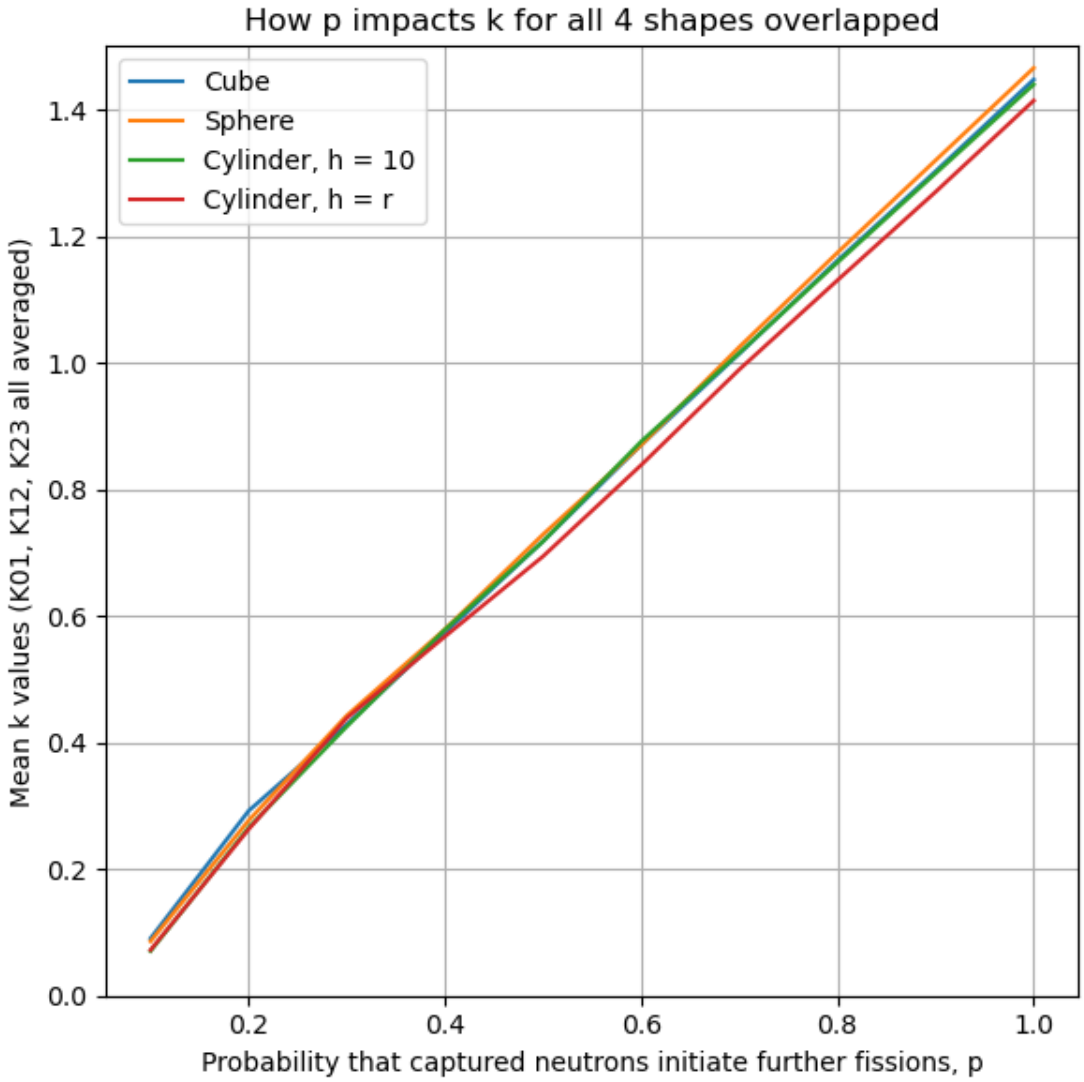


Figure 2: Plot showing how changing the probability of neutrons causing more fissions affects different k values for different shapes.

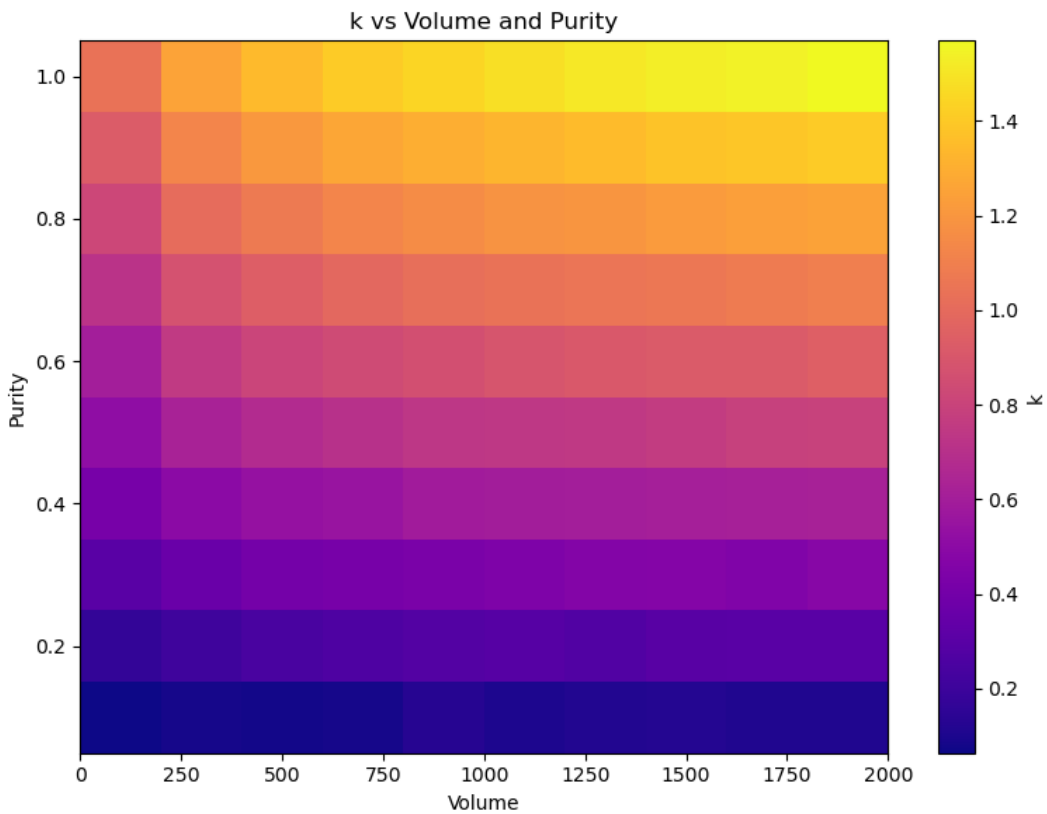
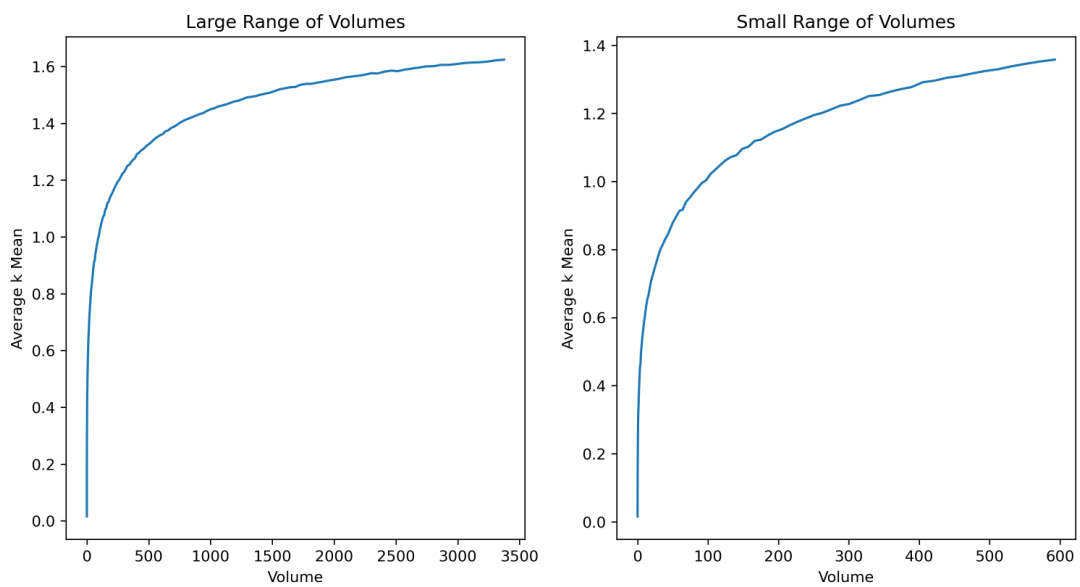
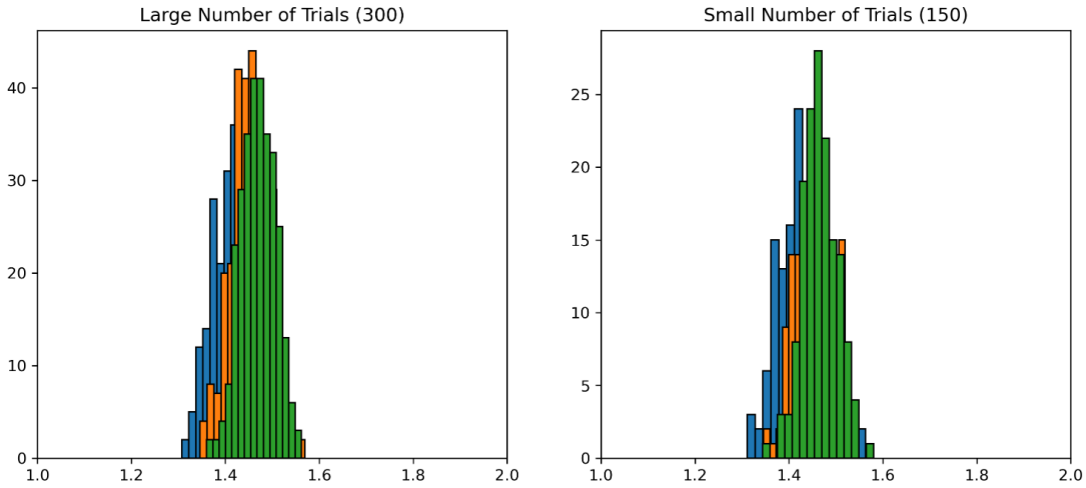


Figure 3: Heatmap showing relationship between k, Purity (p) and Volume (v) for a sphere




Large Number of Trials (300)
Small Number of Trials (150)
0
10
20
30
40
5
15
25
1.0
1.2
1.4
1.6
1.8
2.0